COMSATS University Islamabad, *Lahore Campus*

Department of Computer Engineering

Assignment 2 – FALL 2026

Course Title:	Digital System Design			Course Code:	CPE344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Engr. M Babar Ali			Program Name:	BSc in Computer Engineering		
Semester:	7th	Batch:		FA23	Section:		A/B
Chapters/ Topics:	Ch 4	**Due Date:**			**Maximum Marks:**		**30**
Student's Name:			Reg. No.	CUI/ - - /LHR			

Important Instructions / Guidelines:

- Write your name and Student registration number clearly on the separate title page (cover page of assignment).
- Answer all the questions.
- Do not give multiple answers to a question.
- Submit handwritten assignment on loose pages. Do not use pulled/ripped out papers from notebook.
- Do not use more than one ink color. Use only blue or black fountain pen or ballpoint pen.
- Do show all intermediate steps while solving a question.
- Plagiarism of work from any source will be penalized and may result in zero out of 30 marks in the assessment.

Question 1: **[CLO1, PLO3 – C5]** **(Marks 10)**

Design a digital system for a 4-bit digital volume control system for an audio device that uses a counter to adjust the volume level. The system has four control inputs: `up` to increase the volume, `down` to decrease the volume, `mute` to synchronously set the volume level to minimum (00000000), and `full` to synchronously set it to maximum (11111111). If more than one control input is activated simultaneously, the volume level must remain unchanged. Use an 8-bit parallel-load register as the main building block, along with the necessary combinational logic to implement the design.

Question 2: **[CLO1, PLO3 – C5]** **(Marks 10)**

Design a digital temperature monitoring system for an industrial environment. The system activates an alarm when the average of four consecutive temperature samples is greater than or equal to a user-defined warning threshold. The system has the following inputs and outputs: `CT` is 32-bit unsigned input representing the current temperature sample. `WT` is a 32-bit unsigned input representing the warning threshold. The `clr` is a 1-bit input that disables the alarm and stops the sampling process when set to 1 and `alarm` is a 1-bit output that becomes 1 when the average temperature meets or exceeds the threshold.

Question 3: **[CLO1, PLO3 – C5]** **(Marks 10)**

Design a 4-bit digital score counter for a game system with three control inputs: `cnt` enables the counter to increment the score, `clear` synchronously resets the score to 0000, and `set` synchronously sets the score to 1111. The counter should update its value on the active clock edge.

(a) Implement the design using a 4-bit parallel-load register as the main building block.
(b) Implement the design directly using flip-flops and multiplexers, following the register design process described in Section 4.2.

Good Luck ☺